

Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 3 · SLO 2 — Solved Worksheet

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Unit “ 3 : Session “ 3 : SLO “ 2 | Stack Applications “” Match the Following

■ I. Answers

- A. Expression Evaluation “” 8. Converts infix to postfix or prefix
- B. Reverse Characters “” 3. Characters are popped in reverse order
- C. Syntax Parsing “” 9. Helps in checking for balanced symbols
- D. Backtracking “” 6. Used in solving mazes or puzzles
- E. Undo Mechanism “” 7. Maintains previous states of actions
- F. Function Call Management “” 4. Uses call stack to manage recursion
- G. Parenthesis Checking “” 5. Verifies matching of open/close brackets
- H. Browser History “” 1. Navigates forward and backward
- I. Reversing a String “” 10. Uses stack
- J. Tower of Hanoi “” 2. Uses recursion implemented via stack